

InventOR: A Human-Reviewed, Prompt-Configured Large Language Model Decision Support System for Material-Level Inventory Control

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Abstract

Material planners must set a reorder point, a safety stock and an order quantity for thousands of plant-material pairs, usually under ERP parameters that were last revised long ago. We define and evaluate *InventOR*, a decision support system (DSS) for this recurring decision. For each material, a large language model (LLM) configured only through a prompt, without training or fine-tuning, proposes a reorder-point/order-quantity (r, Q) policy with a written justification and an escalation flag. Deterministic software gates the proposal against minimum order quantities and storage limits and replays it on held-out demand before the planner, who remains the decision-maker, reviews it. We derive six design requirements and evaluate the artifact on 365 plant-material pairs from three plants of an industrial manufacturer, using three independent LLM runs, seven reference arms and an ablation that withholds the ERP planning parameters. In a hard-limit storage replay, the LLM policies cost about 11% less than ERP-derived parameters at a mean fill rate 0.1–0.4 points lower, but 6%, 8% and 18% more than

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history-only, Croston/SBA and empirical-quantile (r, Q) baselines. The saving comes from carrying 13–17% less inventory with 11–20% more stockout days; when shortages are priced at one unit cost, ERP-derived parameters are cheaper and a simulation-tuned policy is cheapest. Static storage checks are insufficient: simulated trajectories exceed storage limits for 53–66% of pairs. InventOR is a checkable, human-reviewed candidate generator, not an optimizer that beats data-driven baselines; benefits to planners are untested. We contribute design principles, review rules and a research agenda.

Keywords: Decision Support Systems, Large Language Models, Generative Artificial Intelligence, Inventory Control, Human–AI Decision Making, Design Science, Operations Research

1. Introduction

Replenishment parameters in manufacturing are set per plant-material pair: a reorder point that triggers an order, a safety stock that absorbs demand and lead-time variability, and an order quantity. In large ERP systems these values often remain at levels set years earlier, while demand patterns, supplier performance and storage constraints change. Revisiting them materially requires analysts who can fit demand and lead-time statistics, apply inventory theory, respect minimum order quantities (MOQs) and storage limits, and explain the result to planners. That scarce skill is exactly what a conversational large language model (LLM) with a code interpreter appears to offer, and a rapidly growing literature explores such tools for supply-chain and operations decisions [1, 2, 3, 4, 5].

The promise is real, but so are three problems that limit its use in practice. First, LLM outputs can be unfaithful or fabricated [6], so a proposed policy must be treated as a candidate and not a decision. Second, claims of

16 superiority over ERP parameters are hard to interpret when the LLM and
 17 the comparators are given different information; an LLM that sees a plant’s
 18 ERP reference values and is compared with an OR policy that does not, or
 19 the reverse, tells us little about the LLM. Third, inventory policies live inside
 20 physical limits: a storage rule that holds when the policy is written down
 21 can be violated as stock builds up over time.

22 This paper treats the problem as one of decision support system (DSS)
 23 design. We define *InventOR* as a human-reviewed, model-driven DSS for
 24 the recurring decision of setting replenishment parameters per plant-material
 25 pair (Section 2). A planner remains the decision-maker. The system supplies,
 26 for each material, a candidate (r, Q) policy proposed by an enterprise LLM
 27 application that is configured only through a prompt and receives no project-
 28 specific training or fine-tuning, together with a justification, an escalation
 29 flag and a counterfactual simulation; deterministic software parses, gates
 30 and simulates every proposal before the planner reviews it and alone decides
 31 whether any ERP parameter is changed. The pattern is not specific to
 32 inventory: any production-planning decision in which parameters are set per
 33 item and refreshed periodically (lot sizes, planned lead times, buffer levels,
 34 dispatching rules) has the same structure of many semi-structured decisions,
 35 scarce analyst time, hard constraints and a need for accountable evidence.
 36 The design lessons below therefore concern LLM-based decision support for
 37 production and logistics planning in general, and not only inventory control.

38 We follow a design-science logic [7, 8]: we state the problem and its scope,
 39 derive design requirements, build the artifact, and evaluate it against those
 40 requirements on real industrial data. Four research questions structure the
 41 paper.

42 **RQ1. *Design.*** How should an LLM be embedded in a DSS so that its
43 proposals are bounded, verifiable, reproducible and reviewable by a
44 planner?

45 **RQ2. *Efficacy.*** What cost and service do the proposed policies achieve
46 relative to ERP-derived parameters and to OR policies, when the
47 information available to each approach is controlled?

48 **RQ3. *Feasibility and reliability.*** Do the policies respect physical storage
49 limits along simulated inventory trajectories, and how stable are they
50 across independent runs?

51 **RQ4. *Use.*** Under which conditions should a planner trust, escalate or reject
52 a proposal?

53 Our contributions are:

54 **C1. A definition and scope of the DSS** (Section 2): the decision,
55 the decision-maker, the system boundary, the level of automation and
56 six design requirements that make the role of the LLM explicit and
57 bounded.

58 **C2. A DSS architecture and operating rules** (Section 4) that separate
59 generation, deterministic gating, simulation and planner review, with
60 explicit rules for when output is trusted, escalated or rejected, and
61 design principles derived from this application (Section 8).

62 **C3. An information-controlled, trajectory-level evaluation** (Sec-
63 tion 5) with seven reference arms that differ in the information they use
64 and in method (ERP-derived, ERP-informed, normal-approximation,
65 Croston/SBA, empirical-quantile and simulation-tuned policies), an

LLM ablation denied the ERP planning parameters, and a hard-limit storage replay that is the primary result. This separates what the LLM adds from what it is given.

C4. Complete-cohort industrial evidence from three LLM runs on 365 plant-material pairs, with plant-level, regime-level and sensitivity analyses, and managerial and research implications.

Our main findings are as follows. Against ERP-derived parameters the LLM policies are about 11% cheaper at a mean fill rate 0.1–0.4 points lower, but the LLM costs 6%, 8% and 18% more than history-only, Croston/SBA and empirical-quantile (r, Q) baselines at essentially equal service, and its saving over ERP comes from carrying less inventory at the price of more stockout days. Storage limits that hold in a static check are violated along unconstrained trajectories for 53–66% of pairs. The simulation therefore does not show superior optimization by the LLM; it shows that a prompt-configured LLM can produce checkable candidate policies without code or training, whose benefit to planners remains to be tested.

2. Problem, Scope and Definition

2.1. The decision problem

The decision supported is the periodic (re)setting of three replenishment parameters, a reorder point, a safety stock and an order quantity, for each plant-material pair held in an ERP system. In the classification of Gorry and Scott Morton [9] it is an operational-control decision that is partly structured: the calculation is textbook, but the choices of estimator, the treatment of intermittent or trending demand, the plausibility of recorded lead

90 times and the handling of constraints require judgement that is applied in-
 91 consistently when thousands of materials compete for a few analysts' time.
 92 The decision-maker is the material planner or inventory analyst; the deci-
 93 sion is repeated, high-volume and consequential for working capital, service
 94 and storage. Hard constraints are the minimum order quantity (MOQ) and
 95 a physical storage limit. The feedback loop is slow: a changed parameter
 96 shows its effect only in subsequent demand.

97 2.2. Definition and level of automation

98 **Definition.** *InventOR is a human-reviewed, model-driven deci-*
 99 *sion support system that, for one plant-material pair at a time, (i)*
 100 *obtains from a large language model, configured only by a written*
 101 *prompt, a candidate (r, Q) policy with a justification and an es-*
 102 *calation flag; (ii) verifies the candidate deterministically against*
 103 *explicit constraints; (iii) evaluates it by counterfactual simulation*
 104 *against reference policies on held-out demand; and (iv) presents*
 105 *the result to a planner, who alone decides whether ERP parame-*
 106 *ters are changed.*

107 In the classical DSS architecture of data, model and dialog components
 108 [10, 11], the LLM and the simulator form the model component, the material-
 109 scoped data extract and the prompt form the data and knowledge compo-
 110 nents, and the review interface is the dialog component (Section 4.1). In
 111 terms of the levels of automation of Parasuraman et al. [12], InventOR op-
 112 erates at a low-to-intermediate level: it selects and suggests one alternative
 113 and leaves execution to the human. This is a design decision, not a limita-
 114 tion: acceptance of automated suggestions is accompanied by complacency

115 and automation bias [13], which the review rules of Section 4.3 are meant to
116 counter.

117 *2.3. Scope*

118 InventOR covers the periodic review of reorder point, safety stock and
119 order quantity for one plant-material pair at a time (single item, single ech-
120 elon, continuous review, lost sales, (r, Q) family). Order release, supplier
121 selection, pricing, forecasting-model development, multi-item coupling and
122 multi-echelon networks are out of scope. The LLM is a candidate generator
123 with a justification and an escalation flag; it does not forecast, place orders,
124 run the simulation or take the final decision, which stays with the planner.
125 The evaluation is an ex-post simulation on real post-cutoff demand with
126 information-controlled comparison, hard storage-limit replay and repeated
127 runs; field trials, planner workload, decision quality and user acceptance are
128 outside it.

129 *2.4. Design requirements*

130 Four sources motivate the requirements: the known unreliability of LLM
131 outputs [6], guidance on human–AI interaction and accountability [14, 15],
132 the run-to-run nondeterminism of LLM inference [16], and the practical con-
133 straints of inventory control. They yield six requirements (Table 1), each
134 tied to a design feature (Section 4) and an evaluation (Section 6).

135 *2.5. Research approach*

136 We follow the design-science research process of problem identification,
137 requirements, artifact design and evaluation [7, 8]. In the evaluation frame-
138 work of Venable et al. [17], our evaluation is artificial and ex post: it uses
139 real industrial data and a faithful simulation but no live users. It therefore

Table 1: Design requirements, the design features that address them and where they are evaluated.

ID	Requirement	Design feature	Evaluation
DR1	Bounded generation: the LLM proposes and has no authority	Narrow candidate-generator role; prohibited methods; no order placement	Sections 6.1, 6.3
DR2	Verifiable feasibility along time, not only at a point	Deterministic gate; hard-limit trajectory replay	Sections 6.1, 6.5
DR3	Counterfactual evidence for the reviewer	Held-out simulation against reference arms	Sections 6.2, 6.8
DR4	Information transparency: comparisons control what each arm sees	Seven reference arms; LLM ablation without ERP parameters	Sections 6.3, 6.4
DR5	Reproducibility and accountability of the configuration	Frozen prompt file; per-call log; repeated runs	Section 6.7
DR6	Human control with explicit escalation	Trust, escalate and reject rules; escalation flag	Sections 4.3, 6.6

140 establishes technical efficacy, feasibility and reliability; it does not establish
 141 usability, adoption or improved planner decisions, which we list as limitations
 142 and research directions.

143 3. Related Work and Positioning

144 *Decision support systems and human-AI decision making..* DSS research
 145 has long distinguished data-, model- and knowledge-driven systems and has
 146 stressed that the value of a DSS lies in improving the judgement of a human
 147 decision-maker rather than in replacing it [9, 10, 11]. LLMs add a new kind
 148 of model component: it is configured through language, produces free-text
 149 justification and is not deterministic. Research on human-automation in-
 150 teraction shows that the way suggestions are scoped, presented and verified
 151 determines whether users rely on them appropriately or over-rely on them
 152 [12, 13], and guidelines for human-AI interaction call for explicit scoping, ex-
 153 planation and correction mechanisms [14]. Our contribution to this stream is
 154 a concrete, evaluated instance of an LLM-based DSS for an operational pro-
 155 duction decision, with design requirements that are tied to evidence. Recent

156 DSS work derives design principles for LLM-based assistants [18], reports de-
 157 sign and pilot implementations of AI-enabled decision support in operations
 158 [19], and shows that the uptake of algorithmic advice depends on trans-
 159 parency and on the user’s metacognition [20, 21]; evaluations of data-driven
 160 forecasting systems in practice likewise stress lessons beyond accuracy [22].
 161 Model-driven decision support is also known to shape how groups handle
 162 disagreement [23]. Human override of algorithmic safety-stock advice is a
 163 documented risk in inventory settings [24], which is why InventOR keeps the
 164 planner as decision-maker and makes the evidence for each proposal explicit.

165 *LLMs in operations and supply-chain decisions..* Reviews and frameworks
 166 map the opportunities for generative AI in supply chains [1, 2, 25, 26, 3, 4, 27]
 167 and stress that adoption depends on assigned human roles, data quality and
 168 governance [5, 28, 29, 30, 31]. Case evidence from manufacturing planning
 169 is still thin: examples include a fine-tuned LLM for digital-manufacturing
 170 optimization [32] and a transformer trained to set order timing and quan-
 171 tity, benchmarked against (r, Q) [33]; experiments with managers also show
 172 that ChatGPT reproduces human decision biases [34]. A recent conceptual
 173 line proposes agentic digital twins that combine AI agents with model-based
 174 simulation and optimization to support decisions [35], and an LLM-plus-
 175 digital-twin method has been built for closed-loop supply-chain operations
 176 [36]. Our decision support system follows the same logic at a smaller scale:
 177 an LLM proposes and a simulation model checks. Most technical LLM work
 178 focuses on translating natural-language problem statements into optimiza-
 179 tion models and solver code [37, 38, 39, 40], on multi-agent control and
 180 consensus-seeking in inventory flows [41, 42], on extracting supply-network
 181 structure from text [43, 44], on disruption modelling [45], or on evaluating

LLMs on classical operations-research tasks such as vehicle routing [46]. The closest LLM study to ours uses an LLM as a forecasting copilot for spare-parts demand, combined with simulation, and reports high variability for irregular demand that can be reduced by averaging repeated queries and enriching prompts [47]. We share the concern about run-to-run variability and test it directly (Section 6.7), but differ in that the LLM proposes a complete replenishment policy rather than a forecast.

Data-driven inventory control without LLMs. Learning-based replenishment is well established, including data-driven policies learned from sequentially observed non-stationary demand [48] and forecasts augmented with leading indicators [49], and heuristics remain competitive in constrained settings such as perishable inventory [50]. Deep reinforcement learning has been applied to omnichannel and multi-echelon systems [51, 52, 53] and aerospace replenishment [54]; deep-learning forecasting has been integrated with replenishment [55]; contextual distributionally robust optimization uses side information to set order quantities [56]; and distribution-free online algorithms handle capacitated warehouses under uncertain price and demand [57]. Robust inventory control under demand and lead-time uncertainty has a longer history [58, 59, 60], and lead-time variability and dependence are known to propagate through ordering decisions [61], while safety stock and safety lead time are alternative buffers against unreliability [62]. The closest operational neighbour is a trained data-driven framework that jointly sets replenishment parameters, safety stock and safety lead time on real automotive data [63]. Reinforcement learning has also been combined with demand forecasting to optimize order points [64]. These methods require model training, historical scenario generation or tuned

hyperparameters for each setting; InventOR requires none, at the price of an LLM-based generator whose internal logic cannot be inspected beyond its written justification.

Intermittent and spare-parts demand. Material histories in manufacturing are frequently intermittent or lumpy. The statistical literature offers specialized estimators and classification schemes [65, 66, 67, 68, 69], and recent studies develop feature-based forecast combinations and temporal aggregation for intermittent demand [70, 71], forecasting for spare parts across product life-cycle phases [72], and integrated forecasting and planning or multi-echelon stocking models for repairable and spare parts [73, 74]. Related industrial studies examine how manufacturers identify and select safety buffers [75]. Our regime analysis (Section 6.6) shows where an LLM that is forbidden from using specialized intermittent-demand estimators behaves differently on lumpy materials.

In summary, InventOR is not a new inventory model and does not claim to beat trained or optimized methods. It contributes (i) a DSS design in which an off-the-shelf LLM is a bounded candidate generator inside a reviewed process, (ii) an evaluation that controls the information available to each arm, and (iii) industrial evidence on cost, service and storage feasibility. The (r, Q) policy it uses is the standard workhorse of continuous-review control [76, 77, 78].

4. The InventOR Decision Support System

4.1. Architecture and components

Following the classical DSS components [10, 11], InventOR has a *data* component (a material-scoped extract of daily demand, lead times, working-

233 day flags, cost and constraint parameters and ERP reference values; the fields
 234 supplied are controlled and logged; DR4), a *knowledge* component (the sys-
 235 tem prompt file defining the task, the prohibited methods, the storage rules
 236 and the output schema, Supplementary File S1; DR1, DR5), a *model* com-
 237 ponent split into a generator (an enterprise LLM application with a code
 238 interpreter, configured only by the prompt, proposing one (r, Q) policy with
 239 a justification and an escalation flag; DR1) and a deterministic verifier and
 240 simulator (parser, gate, held-out simulation with hard storage-limit replay
 241 and reference arms; DR2, DR3), a *dialog* component (policy, justification,
 242 flag and simulated comparison shown to the planner under the trust, es-
 243 calate and reject rules; DR6) and a *governance* layer (frozen prompt file,
 244 per-call log of input hash, time, session and raw response, repeated runs;
 245 DR5). The generator is the only component whose internal logic is not in-
 246 spectable; it is surrounded by deterministic and auditable components. In
 247 the evaluated prototype the dialog component is the per-pair output record
 248 and the comparison tables; an interactive planner interface was not built or
 249 evaluated.

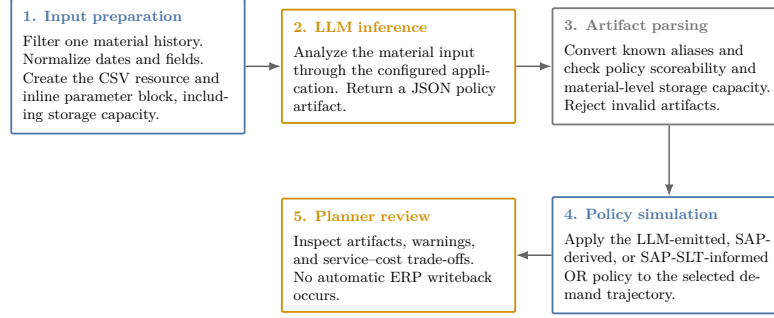
250 4.2. Operation in four steps

251 A planner’s question is: *given what this material has done in the past,*
 252 *how much safety stock should I hold, at what stock level should I reorder,*
 253 *and how much should I order?* InventOR answers this question in four steps
 254 (Figure 1).

255 **Step 1. Generate.** The LLM, accessed through an enterprise application
 256 with a code interpreter, receives one material’s history (daily de-
 257 mand, realized lead times, working-day flags) and a short param-
 258 eter block (unit cost, holding rate, ordering cost, MOQ, service

InventOR prototype workflow

Executed material-level path from prepared history to planner-reviewable simulation output.



Deterministic components reproduce simulation results from stored artifacts; LLM inference itself may vary.

Figure 1: InventOR workflow: material data are prepared, an LLM proposes a policy, deterministic software parses, checks and simulates it, and a planner reviews the result.

target, storage limit, and current ERP safety stock and safety lead time for comparison). A written prompt (the only configuration; no weights are changed) [79] instructs it to compute demand and lead-time statistics, classify the demand pattern, apply an economic-order-quantity (EOQ) argument and a normal-approximation safety stock, respect the storage limit, and report a structured policy together with a short justification and escalation flag.

Step 2. Gate. Deterministic code extracts the policy, rejects malformed or infeasible outputs (missing or non-numeric fields, non-positive reorder point or order quantity, negative safety stock, reorder point below safety stock, safety stock plus order quantity above the storage limit) and retries the generation up to three times. The backtest applies one further admission check, that the order quantity is not below the MOQ.

Step 3. Simulate. Each accepted policy is replayed on the material's de-

mand observed *after* the data cutoff, a held-out evaluation in the sense of out-of-sample forecast testing [80, 81], together with the comparator policies, producing holding and ordering cost, fill rate and stockout days. This plays the role that model-based simulation plays in digital-twin decision support [35], at the scale of a single policy.

Step 4. Review. A planner sees the policy, the LLM’s justification and flag, and the simulated comparison. Only the planner changes ERP parameters.

The role of the LLM is deliberately narrow. It is a *candidate generator* whose output is never trusted without the gate, and it is not used to forecast, to place orders, or to run the simulation. The prompt prohibits Monte Carlo simulation, distribution fitting, machine-learning libraries and specialized intermittent-demand estimators, so that the policy rests on transparent statistics a planner can reproduce.

4.3. Trust, review and reject rules

We propose three explicit rules for planners, in line with guidelines for human–AI interaction and with internal-auditing practice for AI systems [14, 15], and motivated by the evidence in Section 6. The rules and the 0.5-point threshold are design proposals that we did not evaluate with planners; with the baselines used here the trust rule would be met by only a minority of pairs (the LLM is cheaper than the history-only OR in 74–77 of 346), which is itself a finding about the LLM.

Trust (apply after review) when the proposal passes the gate, the simulated trajectory respects the storage limit when orders are truncated at it,

300 and the simulated cost is not above that of the best history-only OR ref-
301 erence (Croston/SBA or empirical quantile) with a fill rate not more than
302 about 0.5 points lower.

303 **Escalate** when the LLM raises an escalation flag, the material is lumpy or
304 has strongly seasonal or trending demand, the plant or calendar data were
305 repaired, or the policy differs strongly between repeated generations.

306 **Reject** when the proposal fails the gate after the retry budget, exceeds the
307 storage limit in the hard-limit replay, or is dominated (higher cost and lower
308 fill) by a history-only reference.

309 *4.4. Provenance and reproducibility of the configuration*

310 The system prompt, which fully defines the task, the forbidden methods,
311 the capacity rules and the output schema, is a plain-text file that is deployed
312 unchanged in the enterprise application; it is distributed with the paper
313 (Supplementary File S1) and the code repository. Every call logs the SHA-
314 256 of its input data, the UTC time, the application session identifier and
315 the full raw response. The platform does not expose the underlying model
316 identifier or sampling settings to the client; the evaluated model (Anthropic
317 Claude Sonnet 4.6, with a code interpreter) is therefore declared by the
318 operator and is not machine-verifiable. We report this limitation openly and
319 base the reproducibility argument on the prompt file, the per-call logs and
320 three independent runs (Section 6.7) rather than on a model checkpoint.

321 5. Application and Evaluation Design

322 5.1. Evaluation approach

323 The evaluation addresses each design requirement in turn (Table 1):
324 whether the pipeline returns valid, feasible policies (DR1, DR2); what cost
325 and service the policies achieve against reference arms on held-out demand
326 (DR3); how much of that result is due to the information given to the LLM
327 rather than to its reasoning (DR4); whether storage limits hold along the
328 simulated trajectory (DR2); how stable policies are across independent runs
329 (DR5); and where proposals should be escalated (DR6). The primary basis
330 is the hard-limit replay with working-day receipts and no shortage penalty;
331 sensitivity analyses vary these choices. The evaluation is ex post and uses
332 no live users (Section 2.5).

333 5.2. Industrial application and data

334 The application is a pilot at a large manufacturer. The dataset contains
335 233,078 daily records for 411 plant-material pairs from three plants (A, B
336 and C), with a data cutoff on 10 March 2026 and a post-cutoff horizon
337 observed to 27 August 2026. Of the 411 pairs, 365 are eligible for scoring
338 (234 in Plant A, 65 in Plant B, 66 in Plant C) and are sent to the LLM. For
339 19 pairs (all in Plant A) the supplied MOQ plus the reference safety stock
340 exceeds the storage limit, so no comparator can satisfy both rules; these are
341 excluded from outcome comparison, leaving a *common-feasible cohort* of 346
342 pairs (215, 65 and 66). All 365 pairs receive an LLM policy. Demand is
343 concentrated: the five largest materials account for 40.3% of demand and
344 the ten largest for 57.8%. The source working-day flag of Plant C is all zero;
345 we reconstruct it from the same-date majority of Plants A and B (ties count

as working days) and test the effect of this repair in Section 6.8. Such data-quality problems are a known threat to analytics-based decision support [82] and, for inventory systems, to record accuracy [83, 84, 85].

Each pair is sent as a material-scoped CSV with daily demand, recorded lead times, a working-day flag, the minimum order quantity, unit cost, holding-cost rate, ordering cost, service-level target, the storage limit, and two *ERP planning parameters* (current safety stock and safety lead time), which are withheld in the ablation arm. The CSV also carries supplier and lane context fields that the prompt does not use.

5.3. Reference arms and information content

The LLM and the classical comparators do not necessarily use the same information, and this matters for the interpretation of any cost difference, because ERP safety stock and safety lead time embed accumulated planner judgement, and safety stock and safety lead time are alternative ways of buffering unreliability [62]. Planned lead times can also drift upward when they are adjusted dynamically (the “lead time syndrome” [86]), which would inflate any comparator that adds them to the lead time. We therefore construct arms that make the information explicit (Table 2).

Three further arms use no ERP input and represent stronger data-driven practice than the history-only OR. The *Croston/SBA OR* replaces the mean daily demand by the Croston estimate with the Syntetos–Boylan bias correction [65, 66] (smoothing constant 0.1, working-day demand history), uses the root mean squared one-step forecast error as demand spread and the history-only lead-time moments, which suits intermittent demand [67]. The *empirical-quantile OR* sets the reorder point to the service-level quantile of the historical lead-time demand, computed from rolling sums over the

Table 2: Comparison arms and the information each uses. All arms respect MOQ and are projected to the storage limit when possible.

Arm	Parameters from	ERP used	inputs	Purpose
SAP-derived	ERP safety stock; EOQ order quantity	Safety stock		ERP-derived reference (not observed practice)
SLT-informed OR	Stationary (r, Q) with ERP safety lead time added to the lead time	Safety lead time		OR policy using the ERP safety lead time
ERP-floor OR	SLT-informed OR with safety stock floored at the ERP value	Both		OR arm matched to the LLM’s full information
History-only OR	Stationary (r, Q) from demand and lead-time history, mean lead time	None		ERP-independent baseline
Croston/SBA OR	(r, Q) with the Syntetos–Boylan-approximated Croston demand rate and its forecast-error spread, mean lead time	None		Intermittent-demand baseline
Empirical-quantile OR	(r, Q) with the reorder point set to the service-level quantile of historical rolling lead-time demand	None		Distribution-free baseline
Simulation-tuned (r, Q)	Reserve factor and order-quantity multiple chosen by simulation on the last 250 working days before the cutoff	None		Trained data-driven baseline
LLM (runs 1–3)	LLM proposal with all CSV fields (Section 5.2)	Both		Main treatment
LLM without ERP	LLM with ERP safety stock and safety lead time set to “not supplied”	None		Ablation: what the LLM does when it cannot anchor on ERP

372 rounded mean lead time in working days, and needs no distributional as-
 373 sumption. The *simulation-tuned* (r, Q) policy is a learned baseline: from a
 374 grid of reserve factors and order-quantity multiples around the history-only
 375 policy it selects the cheapest candidate whose fill rate on the last 250 work-
 376 ing days *before the cutoff* meets the service target (or the best-fill candidate
 377 if none does). No post-cutoff data are used in any of these arms. We do not
 378 implement a deep reinforcement-learning baseline [52, 51]; the simulation-
 379 tuned policy is the trained comparator, and Section 8 lists neural baselines

380 as a limitation.

381 5.4. Simulation, capacity replay and metrics

382 Each pair is simulated independently on the post-cutoff horizon as a
383 single-item, single-echelon continuous-review (r, Q) system [76, 87, 88, 78, 77]
384 with lost sales [89], in the spirit of simulation-based experiments on supply
385 and timing uncertainty [90], a shared opening stock, an empty pipeline and
386 arrivals after the rounded mean lead time in working days (Appendix Ap-
387 pendix A). MOQs are treated as hard order-size constraints, as imposed
388 by suppliers [91], and storage limits as capacity bounds on the replenish-
389 ment decision [57]. The primary metrics are the mean material-horizon
390 holding-plus-ordering cost, the mean material-level fill rate, the aggregate
391 (volume-weighted) fill rate and the number of stockout days. We report
392 paired differences with 10,000-sample bootstrap 95% intervals.

393 A static check $SS + Q \leq \text{max_storage_units}$ does not show that storage
394 limits are respected as the inventory evolves, because the reorder point in-
395 cludes lead-time demand and inventory on order can exceed the limit when
396 it arrives. We therefore record, for every policy and every simulated day, the
397 post-receipt on-hand stock and report the share of pairs and days above the
398 limit (the *unconstrained* simulation). We then *replay* every policy with each
399 order truncated to the storage limit minus the current inventory position
400 (the *hard-limit replay*). The hard-limit replay is our primary evaluation.

401 5.5. Repeated runs and sensitivities

402 To examine run-to-run variation, the entire cohort is generated three
403 times (runs 1, 2 and 3) with the same prompt file and the same data. Run 1
404 serves as the reference for tabulated results unless stated otherwise; runs 2

and 3 provide replicates. Sensitivity analyses vary the working-day reconstruction (majority, any-plant, strict-all, exclusion of Plant C and calendar-day arrivals), the shortage penalty (zero, one and five times unit cost).

6. Results

6.1. Completion and policy characteristics

All four runs (three with the full input and one ablation, Section 6.4) return a valid, storage-feasible (r, Q) policy for all 365 pairs after at most three attempts per pair. The validity gate rejects non-positive reorder points and order quantities, negative safety stocks and reorder points below the safety stock. The ERP safety stock is zero for 198 of the 365 pairs (188 of the 346 in the cohort below), so for a majority of pairs the SAP-derived arm holds no safety stock at all.

6.2. Headline comparison with ERP-derived parameters

Unless stated otherwise, all results from here on use the hard-limit replay (every order truncated at the storage limit) on the 346-pair common-feasible cohort, with working-day receipts and no shortage penalty; Section 6.5 reports the unconstrained simulation for comparison. Table 3 compares all arms, including the LLM cost relative to each arm in the three runs (positive values mean that the LLM costs more). In run 1 the LLM policies cost 11.4% less than SAP-derived policies and 30.5% less than the SLT-informed OR comparator (mean 2,928 versus 3,305 and 4,211). The mean fill rate is 0.18 points lower than for SAP-derived (98.11% versus 98.30%; bootstrap interval -0.54 to $+0.10$) and 0.11 points lower than for the SLT-informed OR. The mean cost difference against SAP-derived is -377 (95% interval -651 to -154); the LLM is cheaper than SAP-derived in 241 of 346 pairs

and, in 226 pairs, simultaneously no more expensive and no worse in fill. The LLM incurs more stockout days in total (910 versus 804 and 831), and aggregate fill is 78.87% versus 79.34% and 79.15%.

The cost decomposition shows where the saving comes from. Relative to SAP-derived parameters the LLM carries about 17% less average inventory (16,256 versus 19,565 units) and has about 16% lower holding cost, while it places about the same number of orders per year (17.3 versus 17.6). The saving is therefore a trade of inventory for service: the LLM holds less stock and accepts 11–20% more stockout days (runs 3 and 2). It is not the consequence of a better ordering rhythm, and it is not free of service cost. Section 6.8 shows that, once shortages carry a cost, this trade no longer favours the LLM.

Table 3: Hard-limit replay, common-feasible cohort ($n = 346$), working-day receipts, zero shortage penalty. Cost is the mean material-horizon holding-plus-ordering cost; average inventory is in units; orders are per year. The last column gives the LLM cost relative to the arm in runs 1/2/3 (positive: the LLM costs more).

Arm	Mean cost	Holding	Ordering	Avg. inv.	Orders	Mean fill (%)	Aggr. fill (%)	Stockout days	LLM vs arm (%)
ERP-derived (SAP)	3,305	2,172	1,132	19,565	17.6	98.30	79.34	804	−11.4 / −11.3 / −11.4
SLT-informed OR	4,211	2,608	1,603	21,614	24.5	98.23	79.15	831	−30.5 / −30.4 / −30.5
ERP-floor OR	4,248	2,633	1,616	21,937	24.7	98.23	79.15	831	−31.1 / −31.0 / −31.1
History-only OR	2,764	1,725	1,039	15,972	16.3	98.14	78.72	874	+5.9 / +6.0 / +5.9
Croston/SBA OR	2,705	1,665	1,040	15,521	16.3	98.30	78.87	803	+8.2 / +8.4 / +8.3
Empirical-quantile OR	2,489	1,543	946	13,331	14.8	98.35	78.87	747	+17.7 / +17.8 / +17.7
Simulation-tuned (r, Q)	3,266	2,473	793	11,914	12.6	96.05	84.84	1,973	−10.4 / −10.3 / −10.3 [†]
LLM (run 1)	2,928	1,824	1,104	16,256	17.3	98.11	78.87	910	
LLM (run 2)	2,931	1,850	1,081	16,599	17.0	97.89	78.70	968	
LLM (run 3)	2,929	1,841	1,088	16,958	17.0	98.16	78.90	890	
LLM without ERP	3,014	1,902	1,112	16,507	17.3	98.06	78.89	890	−8.8 [†] (vs SAP)

[†] Other comparisons of the ablation are in Section 6.4. [‡] Different operating point (mean fill 96.05%, aggregate fill 84.8%); the paired interval of the cost difference includes zero, and with priced shortages this policy is the cheapest arm (Section 6.8).

6.3. Information asymmetry: what the LLM adds beyond the ERP inputs

The comparison below uses Table 3. Three results stand out.

First, the SLT-informed OR and its ERP-floor variant are the most expensive arms (4,211 and 4,248). The ERP safety lead time inflates their effective lead time, hence their buffers; against them the LLM is about 30–31% cheaper in every run. This comparison therefore mainly measures how

448 heavily ERP inputs over-buffer, not the quality of the LLM.

449 Second, the *history-only OR*, which uses no ERP inputs, has a cost of
450 2,764 at a fill of 98.14%. Against it the LLM costs 5.9–6.0% more in all
451 three runs (run 1: mean difference +164, 95% interval +91 to +240), with
452 fill differences of -0.03 , -0.25 and $+0.01$ points whose intervals include zero
453 in all three runs. The LLM is cheaper than the history-only OR in only 74–
454 77 of 346 pairs. The stronger data-driven arms widen the gap. Against the
455 Croston/SBA OR (2,705, fill 98.30%) the LLM costs 8.2–8.4% more (run 1:
456 mean difference +223, interval +97 to +362; the LLM is cheaper in 91–99
457 pairs), and against the empirical-quantile OR (2,489, fill 98.35%) it costs
458 17.7–17.8% more (mean difference +439, interval +329 to +557; the LLM
459 is cheaper in 44–49 pairs) at a mean fill that is 0.24 points lower. Part of
460 the LLM’s lower cost relative to ERP-derived parameters therefore reflects
461 the weakness of those benchmarks; the LLM does not match a data-driven
462 (r, Q) baseline.

463 The simulation-tuned (r, Q) policy sits at a different operating point. It
464 reaches an aggregate fill of 84.8% with the lowest average inventory (11,914
465 units) but a lower mean fill (96.05%) and many more, smaller stockouts
466 (1,973 stockout days), because its tuning criterion is the fill of the last pre-
467 cutoff year, not the post-cutoff horizon. Against it the LLM costs 10.3–10.4%
468 less in the zero-penalty cost, with a 2.1-point higher mean fill, but the paired
469 interval of the cost difference includes zero ($-2,527$ to $+857$ in run 1). Once
470 shortages are priced it becomes by far the cheapest arm (Section 6.8), so its
471 low zero-penalty ranking is not a sign of a weak baseline.

472 Third, the LLM is nonetheless cheaper than ERP-derived parameters
473 (11.3–11.4%) at a mean fill that is lower by 0.14–0.40 points (the interval
474 excludes zero only in run 2), so it improves on the ERP-derived reference.

475 It does not reach the best classical baselines, which also yield a policy for
 476 every material at lower cost; the evidence therefore does not establish a
 477 cost advantage of the LLM over data-driven alternatives. Whether the LLM
 478 relies on the ERP planning parameters it is given is tested in the ablation of
 479 Section 6.4.

480 6.4. LLM without ERP planning parameters (ablation)

481 In this ablation the LLM receives exactly the same material histories,
 482 cost parameters, MOQ, service target and storage limit as in runs 1–3, but
 483 the ERP safety stock and ERP safety lead time are blanked in the uploaded
 484 CSV and marked “not supplied” in the inline parameter block; the system
 485 prompt file is unchanged. It tests whether the LLM anchors on the ERP
 486 planning parameters it is given, and therefore whether the comparison with
 487 the ERP-informed arms is confounded by information. All 365 pairs returned
 488 a valid, storage-compliant policy.

489 *Policy values.* Withholding the ERP parameters changes the LLM’s poli-
 490 cies only slightly beyond its ordinary run-to-run variation. The median ratios
 491 of safety stock and reorder point to run 1 are both 1.00 (346 common pairs).
 492 The reorder point differs by more than 10% in 54.6% of pairs, against 48.3%
 493 between runs 1 and 3 and 48.0% between runs 2 and 3 with the ERP pa-
 494 rameters supplied; the safety stock differs by more than 10% in 11.0% of
 495 pairs, against 9.8% and 9.5%. The median absolute reorder-point difference
 496 is 128 units, against 90 and 84 between ERP-supplied runs. The mean ratio
 497 is slightly above one (1.04 for safety stock, 1.05 for the reorder point), so the
 498 LLM tends to buffer a little more when it has no ERP anchor.

499 *Outcomes.* In the hard-limit replay the LLM without ERP parameters
 500 has a mean cost of 3,014 at a mean fill of 98.06%, against 2,928–2,931 and

501 97.89–98.16% in runs 1–3. It costs 8.8% less than SAP-derived parameters
 502 (95% interval of the mean difference -566 to -61 ; fill -0.24 points, inter-
 503 val including zero) and 28.4–29.1% less than the SLT-informed and ERP-
 504 floor ORs, but it costs 9.0% more than the history-only OR (interval $+134$
 505 to $+387$; fill -0.09 points; cheaper in 91 of 346 pairs), 11.4% more than
 506 the Croston/SBA OR and 21.1% more than the empirical-quantile OR, and
 507 7.7% less than the simulation-tuned policy at its different operating point.
 508 Withholding the ERP parameters thus raises the LLM’s cost by roughly 3
 509 percentage points relative to the history-only OR and does not reverse any
 510 ranking.

511 *By plant.* The pattern is the same as with ERP inputs: the LLM costs
 512 less than SAP-derived at Plants A (215 pairs, -10.8%) and B (65 pairs,
 513 -21.9%) and more at Plant C (66 pairs, $+41.6\%$ against $+36.2\%$ in run 1),
 514 so the Plant C loss is not caused by reliance on ERP parameters.

515 *Interpretation.* Two conclusions follow. First, the information-
 516 asymmetry concern (that the LLM benefits from ERP inputs) does not
 517 explain the LLM’s performance: with the ERP parameters removed the
 518 LLM keeps most of its advantage over ERP-derived parameters, and the
 519 gap to the history-only OR widens only moderately. Second, the ERP
 520 parameters are of limited use to the LLM; most of its output appears to be
 521 determined by the demand and lead-time histories. One limitation is that
 522 the ablation tests only the two ERP planning fields; the LLM still sees unit
 523 cost, holding rate, ordering cost, MOQ, service target and storage limit,
 524 which all reference arms also use.

525 *6.5. Storage limits along simulated trajectories*

526 All 365 LLM policies satisfy the static rule $SS+Q \leq \text{max_storage_units}$,
527 yet in the unconstrained simulation the post-receipt on-hand stock exceeds
528 the limit for 183 pairs (52.9%) for the LLM (178–183 across the three runs;
529 10.2% of simulated pair-days), 198 pairs (57.2%) for SAP-derived and 230
530 pairs (66.5%) for the SLT-informed OR; no pair starts above the limit.
531 Exceedances arise because inventory on order and lead-time demand add to
532 stock above $SS + Q$, so the static check is necessary but not sufficient for
533 any method. Under the hard-limit replay no arm exceeds the limit by more
534 than one unit. Enforcing it raises the LLM mean cost from 2,850 to 2,928
535 and SAP-derived from 3,184 to 3,305 (SLT-informed OR from 3,852 to
536 4,211), without changing the ranking, and we use the replay as the primary
537 evaluation.

538 *6.6. Plants and demand regimes*

539 The average hides a strong plant effect (Table 4, hard-limit replay, run 1).
540 The LLM costs 14.1% less than SAP-derived in Plant A (cheaper in 179 of
541 215 pairs) and 22.7% less in Plant B (54 of 65), but 36.2% *more* in Plant
542 C, where it is cheaper in only 8 of 66 pairs. The Plant C loss is similar
543 in the other full runs (+32.2% and +31.5%) and larger without the ERP
544 parameters (+41.6%). Plant C is also the plant whose calendar we repaired,
545 but Section 6.8 shows that the loss persists under every working-day recon-
546 struction in which Plant C can be scored. Against the data-driven baselines
547 the LLM costs more in every plant: 7.2%, 3.6% and 6.7% more than the
548 history-only OR in Plants A, B and C; 13.1%, 2.3% and 6.2% more than the
549 Croston/SBA OR (it is cheaper in 26 of 65 Plant B pairs and 20 of 66 Plant
550 C pairs); and 20.8%, 7.7% and 29.5% more than the empirical-quantile OR.

Plant B is where the LLM comes closest. The SLT-informed and history-only ORs coincide in Plant C because the ERP safety lead time is zero for all 66 of its pairs.

By demand regime, the 33 lumpy materials show the largest cost reduction against SAP-derived (-25.4% in run 1; -23.7% to -25.2% in the other runs), whereas the 313 smooth materials show -9.8% at a 0.24-point lower mean fill. The service effect on lumpy materials is inconsistent (mean fill $+0.38$ points in run 1, -0.29 in run 2), the history-only OR costs 9% less than the LLM there at a much lower aggregate fill, and the Croston/SBA OR costs 9.7% less at a higher mean fill (94.55% versus 93.12%); only the empirical-quantile OR is slightly more expensive (1.2%; the LLM is cheaper in 15 of 33 pairs). With 33 materials this is indicative only. By terciles of the coefficient of variation, the LLM costs 15.7% less than SAP-derived at equal fill in the lowest tercile, 7.5% less with a 0.68-point lower mean fill in the middle tercile, and 12.9% less in the highest.

Table 4: Mean simulated cost per pair and mean fill by plant, run 1, hard-limit replay (common-feasible cohort). The SLT-informed and history-only ORs coincide in Plant C.

Plant (n)	LLM cost	SAP-derived cost	SLT-informed OR cost	History-only OR cost	Croston/SBA OR cost	Empirical-quantile OR cost	LLM vs SAP	Cheaper pairs	Fill LLM / SAP (%)
All (346)	2,928	2,365	4,211	2,784	2,765	2,480	-11.4%	141	93.11 / 93.20
A (215)	2,391	2,792	3,779	2,231	2,114	1,980	-14.1%	179	99.18 / 99.27
B (65)	4,916	6,368	7,929	4,746	4,886	4,566	-22.7%	54	96.23 / 96.47
C (66)	2,719	1,897	2,549	2,549	2,561	2,100	+36.2%	8	96.20 / 96.81

6.7. Run-to-run stability

Across the three runs the aggregate results are very similar: the LLM cost relative to SAP-derived is -11.4% , -11.3% and -11.4% in the hard-limit replay, and the distributions of policy values have similar medians (reorder point 1,584, 1,582 and 1,466; order quantity 2,737, 2,777 and 2,693; safety stock 683, 688 and 669; Figure 2). Individual policies, however, vary. Comparing runs 2 and 3 on the 365 pairs, all three policy values are identical for only 36 pairs; every policy resolves to the (r, Q) family, and the median

absolute differences are 81 units for the reorder point, 6 for the order quantity and 3 for the safety stock. The reorder point differs by more than 10% in about 48% of the pairs, and a few pairs vary strongly (for one material the reorder point was 14,882, 14,947 and 29,218 units in the three runs; for two others the safety stock was zero in one run and 660 or 1,075 units in the others). One generation per material is therefore a noisy estimate: aggregate behaviour is reproducible, individual values should be reviewed. Such non-determinism is a documented property of LLMs [16].

6.8. Sensitivity analyses

Plant C calendar.. Plant C’s source working-day flag is all zero, which required reconstruction. We re-scored all policies under four alternatives: any-plant, strict-all, exclusion of Plant C and calendar-day arrivals. Where Plant C can be scored (any-plant and calendar-day arrivals), the LLM remains 31–35% more expensive than SAP-derived there in the three full runs, and the cohort-wide advantage is -10.8% to -11.1% (primary calendar: -11.3% to -11.4%). The strict and exclusion variants (280 pairs) drop Plant C and show a larger advantage (-17.0% to -17.6%). The Plant C result is therefore not an artefact of the calendar repair; one residual risk is that the LLM saw Plant C demand series built on the repaired calendar.

Shortage penalty.. With a positive lost-sales penalty the zero-penalty cost ranking changes. Table 5 gives mean total cost per pair (holding, ordering and shortage cost) in the hard-limit replay for penalties of one and five times the unit cost; the zero-penalty costs are in Table 3. Costs are dominated by the shortage term and the gaps between arms are small in relative terms, so the ordering among the non-tuned arms (differences of 0.1–3.7%) should be read with caution. The simulation-tuned policy is by far the cheapest arm as

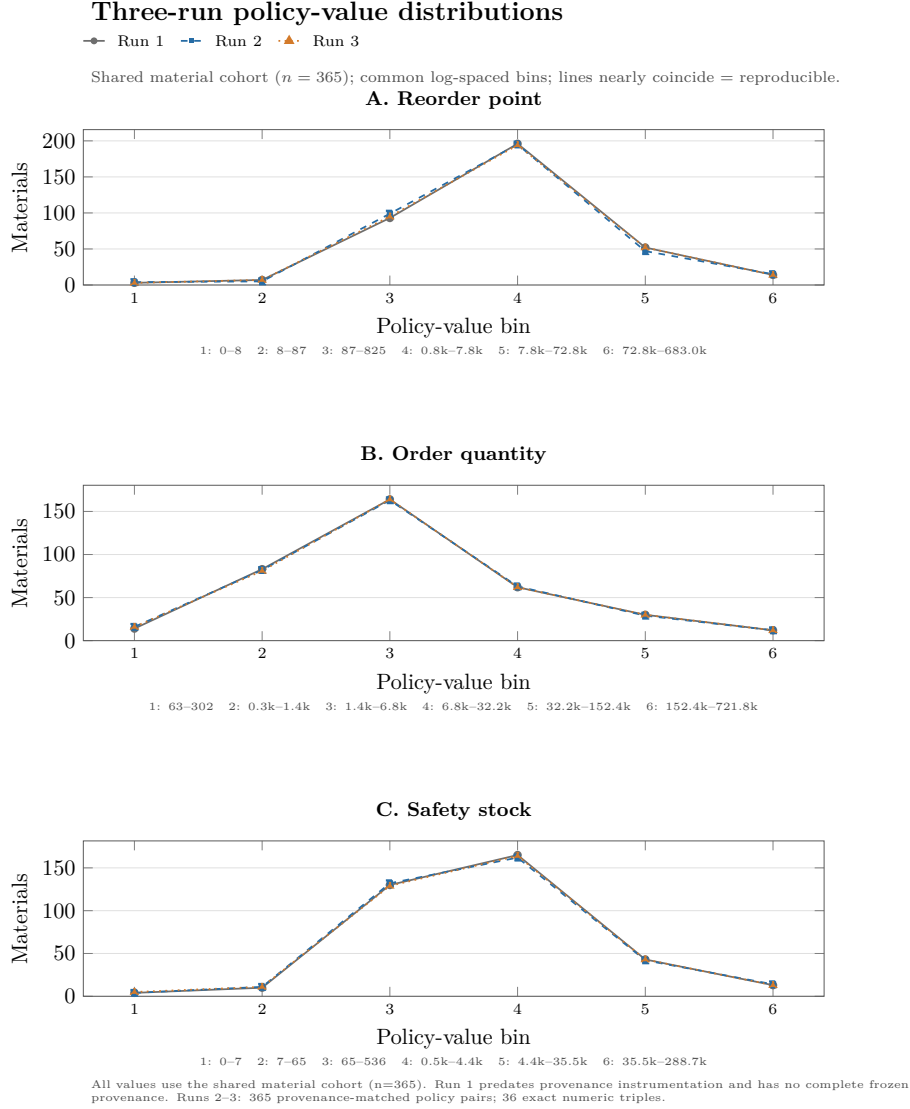


Figure 2: Policy-value distributions across the 365 shared pairs for runs 1, 2 and 3 (materials per common log-spaced bin); similar lines indicate similar aggregate distributions, not identical policies for each material.

600 soon as shortages are priced (1,094,043 at one times the unit cost, about 33%

601 below SAP-derived), because it was selected to meet the service target on

602 the pre-cutoff history. Among the other arms the SAP-derived parameters

are the cheapest, because they carry more inventory and fewer stockouts. Run 1 of the LLM then costs 2.4% more than SAP-derived at both one and five times the unit cost; it is about as expensive as the Croston/SBA OR (1,673,003 versus 1,674,925 at one times the unit cost), and it is cheaper than the SLT-informed OR (1.2% lower at both multiples), the ERP-floor OR, the empirical-quantile OR and the history-only OR (0.8% lower). The LLM’s cost advantage over ERP-derived parameters is therefore specific to the case in which service shortfalls are not charged.

Table 5: Mean total cost per pair in the hard-limit replay (346 pairs) when lost sales are charged at a multiple of the unit cost.

Arm	1×	5×
SAP-derived	1,634,136	8,157,462
SLT-informed OR	1,694,183	8,454,072
ERP-floor OR	1,694,220	8,454,109
History-only OR	1,687,191	8,424,899
Croston/SBA OR	1,674,925	8,363,806
Empirical-quantile OR	1,693,784	8,458,966
Simulation-tuned (r, Q)	1,094,043	5,457,151
LLM run 1	1,673,003	8,353,306
LLM run 2	1,683,280	8,404,676
LLM run 3	1,671,759	8,347,082
LLM no ERP	1,689,961	8,437,748

7. Managerial Insights

- 1. Position the tool as a reviewed decision support system, not an autonomous policy engine.** Single generations differ at the material level (Section 6.7). The gate-simulate-review structure (Section 4.3) yields a candidate policy for every material with a justification and leaves the decision to the planner; whether this helps planners in practice has not been tested.
- 2. Benchmark against data-driven OR baselines before claiming value.** ERP-derived parameters and ERP-informed OR policies are weak

620 benchmarks: safety stocks are often stale or zero and safety lead times in-
 621 flate the effective lead time. A planning team that evaluates an LLM aid
 622 only against ERP will overestimate its benefit. History-only (r, Q) base-
 623 lines (normal approximation, Croston/SBA, empirical lead-time-demand
 624 quantile) are cheap to compute and the LLM cost 6%, 8% and 18% more
 625 than they did in our data, respectively; a policy tuned by simulation on
 626 the pre-cutoff history was cheapest once shortages were priced.

627 **3. Where the LLM was closer to the baselines.** Against ERP-derived
 628 parameters the LLM was cheaper in Plants A and B (14–23%) and on the
 629 33 lumpy materials (about 25%, with mixed service effects). In Plant C,
 630 where the LLM lost, conditions such as repaired calendars, small samples
 631 or atypical lead-time behaviour may call for escalation; we did not test
 632 this.

633 **4. Verify storage along the trajectory.** The static rule $SS + Q \leq \text{limit}$
 634 cannot protect a warehouse: 53–66% of pairs breach the limit in simu-
 635 lation under each of the three arms checked. Operations should enforce
 636 a cap in the order-release logic (order-up-to or truncation), or treat the
 637 limit as a design requirement in the policy.

638 **5. Price shortages explicitly.** Without a shortage penalty the cost rank-
 639 ing rewards under-stocking. With a penalty of one unit cost the LLM’s
 640 cost advantage over SAP disappears (it is about 2.4% more expensive).
 641 Planners should set service targets and penalties before reading the cost
 642 comparison.

643 **6. Freeze and log the configuration.** Treat the system prompt as a
 644 controlled file, log every call (input hash, time, session, raw response)

645 and re-run before changing the prompt. Our reproducibility argument
646 rests on this log, not on the platform’s model identifier, which the client
647 cannot see.

648 **7. Cost and latency.** Each LLM generation took several minutes, whereas
649 the OR baseline takes milliseconds. The LLM is attractive where ana-
650 lyst time and not compute is scarce, but a deployment needs batching
651 and monitoring. Periodic regeneration as demand and lead times change
652 would also support adaptable planning under uncertainty [92].

653 **8. Discussion, Limitations and Research Perspectives**

654 *What the evidence shows..* In a three-plant industrial dataset, a prompt-
655 configured LLM generates feasible, reviewable (r, Q) policies for every ma-
656 terial. In the simulation they cost less than ERP-derived parameters but
657 more than history-only data-driven OR policies (Croston/SBA, empirical-
658 quantile). We therefore do not present InventOR as a method that outper-
659 forms optimization; its claimed role is that of a reviewed candidate generator,
660 and whether that role helps planners is a hypothesis that the simulation can-
661 not test. The information-controlled evaluation is what supports this read-
662 ing: against the ERP-informed arms alone, the LLM would appear much
663 better (-11.4% against SAP and -30.5% against the SLT-informed OR in
664 the hard-limit replay).

665 *Design principles for LLM-based decision support..* The evidence supports
666 six principles that we derive from this application and offer as hypotheses
667 for further design-science work [7]; they are not tested beyond it. **DP1,**
668 **bound the generator:** give the LLM the role of candidate generator with
669 a prompt that fixes the method, the constraints and the output schema, and

no authority to act. **DP2, verify deterministically and along time:** check every output with code, and check feasibility along the trajectory, because point checks of constraints can pass while the constraints are violated over time (Section 6.5). **DP3, show counterfactual evidence:** give the reviewer a simulated comparison with credible references, so that review is based on outcomes and not on the persuasiveness of the justification. **DP4, control information in evaluation:** compare against baselines that use the same information and against a baseline that uses none, and withhold incumbent inputs from the LLM in an ablation, otherwise a gain may merely reflect a weak benchmark (Sections 6.3, 6.4). **DP5, freeze and log:** treat the prompt as a controlled file, log every call and repeat the runs, because individual outputs vary (Section 6.7). **DP6, escalate by rule:** define in advance when proposals are trusted, escalated or rejected, to limit complacency [13] (Section 4.3).

Limitations. (i) The data come from one manufacturer and three plants; generalization is not shown. (ii) The platform does not expose the model identifier or inference settings, so the model is operator-declared. (iii) No planner outcomes were measured. The dialog component is a prototype record, not an evaluated interface, and the claimed benefits of reviewability, auditability and planner support are design intentions that we did not test; a field study with planners is required to measure usability, workload, override behaviour and decision quality. (iv) The simulation uses a deterministic arrival at the rounded mean lead time, excludes observed receipts, corrections and scrap, and starts with an empty pipeline. Lead-time variability is not simulated, which may bias comparisons that depend on safety lead time (for example against the SLT-informed OR). We also did not report

696 whether each arm attained the service target. (v) The primary results use
 697 a zero shortage penalty and standard holding and ordering costs; with a
 698 positive penalty the cost ranking changes. Costs are dominated by a few
 699 high-demand materials and the 346 pairs are not independent draws, so
 700 the bootstrap intervals understate uncertainty. (vi) The (r, Q) policy is a
 701 single-item, single-echelon family; multi-item coupling and shared capacity
 702 are outside the scope. (vii) Run-to-run variation of individual policies is
 703 substantial, and Plant C required calendar reconstruction. (viii) The refer-
 704 ence arms comprise ERP-derived and normal-approximation (r, Q) policies,
 705 Croston/SBA and empirical-quantile (r, Q) policies and a simulation-tuned
 706 (r, Q) policy; we did not include deep reinforcement-learning or neural fore-
 707 casting baselines, so the comparison with the state of the art is incomplete.
 708 The ERP-derived arm is the ERP safety stock combined with our own or-
 709 der quantity and is not an observed record of the plants' actual practice.
 710 (ix) The trust, escalation and rejection rules of Section 4.3 have not been
 711 evaluated.

712 *Research perspectives..*

- 713 1. *Ensembling and consistency.* Aggregating several generations (median or
 714 consensus policies) may reduce the individual-policy noise while retaining
 715 the LLM's justification; the copilot findings for spare parts [47] suggest
 716 this is promising.
- 717 2. *Hybrid generators.* Combining the best data-driven OR policy (Cros-
 718 ton/SBA or empirical quantile) with the LLM (the LLM reviewing and
 719 annotating the OR policy, or choosing among candidates by regime), as
 720 proposed for LLM-OR integration [93], could capture the OR's cost level
 721 and the LLM's flags.

722 3. *Comparison with trained methods.* A benchmark that compares prompt-
 723 based generators with deep reinforcement-learning and other trained poli-
 724 cies [63, 51, 52] on identical data and information would clarify the trade-
 725 off between setup effort and policy quality.

726 4. *Intermittent-demand and richer policy families.* Allowing order-up-to or
 727 (s, S) policies and regime-specific estimators may help the lumpy and
 728 Plant C pairs [68, 69].

729 5. *Planner-in-the-loop field trials.* Prospective studies should compare plan-
 730 ners working with and without the aid on a matched set of materials, and
 731 measure planner time, override rates and realized service and inventory,
 732 which our simulation cannot, and should examine adoption drivers and
 733 barriers with mixed-methods designs [29, 28, 5].

734 6. *Standards for information-controlled evaluation.* Studies of LLM-based
 735 decision support systems should report an ERP-independent baseline, an
 736 ablation that withholds the incumbent’s inputs, trajectory-level feasibil-
 737 ity, and run-to-run variation.

738 9. Conclusion

739 We defined InventOR as a human-reviewed decision support system for a
 740 recurring production-planning decision, the setting of replenishment parame-
 741 ters per plant-material pair, and evaluated it against six design requirements
 742 on a three-plant industrial dataset. A prompt-configured LLM generated fea-
 743 sible, justified (r, Q) policies for every material without training, and a de-
 744 terministic gate, a hard-limit simulation and explicit review rules made these
 745 proposals checkable by a planner. The proposals cost about 11% less than

746 ERP-derived parameters at slightly lower service, but the LLM costs about
 747 6% more than a history-only OR policy, 8% more than a Croston/SBA policy
 748 and 18% more than an empirical-quantile policy; when lost sales are priced
 749 at one unit cost or more, ERP-derived parameters are the cheapest of the
 750 non-tuned arms and a simulation-tuned policy is cheapest overall. Storage
 751 limits that hold in a static check are exceeded along simulated trajectories for
 752 every method, and the LLM loses to ERP at one plant. The evidence there-
 753 fore supports InventOR as a checkable candidate generator inside a reviewed
 754 process, with trajectory-level storage checks, explicit shortage pricing and a
 755 frozen, logged configuration. It does not show that the LLM out-optimizes
 756 a data-driven OR baseline, and the benefit to planners remains to be tested.
 757 The design principles (Section 8) are hypotheses for LLM-based decision
 758 support in settings where parameters are set item by item. Field evidence
 759 on planner behaviour and realized outcomes is the central open step.

760 **Appendix A. Formal Definitions**

761 *Reference calculation..* These are the standard textbook calculations [78, 87].
 762 For pair i , let $\bar{d}_i$ and $\sigma_{D,i}$ be the mean and standard deviation of pre-cutoff
 763 working-day demand, L_i and $\sigma_{L,i}$ the mean and standard deviation of the
 764 lead time in working days, K_i the ordering cost, h_i the annual holding cost
 765 per unit (unit cost times holding rate), z_i the standard normal quantile of
 766 the service target, and 250 the working days per year. The order quantity
 767 is $Q_i = \max\{\text{MOQ}_i, \lceil \sqrt{2K_i \cdot 250\bar{d}_i/h_i} \rceil\}$. The lead-time-demand standard
 768 deviation is $\sigma_{LTD,i} = \sqrt{L_i\sigma_{D,i}^2 + \bar{d}_i^2\sigma_{L,i}^2}$, the safety stock $SS_i = \lceil z_i\sigma_{LTD,i} \rceil$
 769 and the reorder point $R_i = \lceil \bar{d}_iL_i^{\text{eff}} + SS_i \rceil$, where L_i^{eff} is the mean lead time
 770 for the history-only OR and the mean lead time plus the ERP safety lead

time for the SLT-informed OR and the ERP-floor OR. The SAP-derived arm uses the ERP safety stock and the same Q_i . All policies are projected to $SS_i + Q_i \leq \text{max_storage_units}_i$ when $\text{MOQ}_i + SS_i$ fits.

Simulation. On each post-cutoff working day, demand is served from on-hand stock, unmet demand is lost, and an order of size Q_i is placed when the inventory position (on-hand plus on order) is at or below R_i and arrives after the rounded mean lead time in subsequent working observations. Observed receipts, corrections and scrap are ignored. The total cost is $TC_i = h_i \bar{I}_i T / 250 + K_i N_i$, where $\bar{I}_i$ is the mean on-hand stock over the T simulated working days and N_i the number of orders. Fill rate is served divided by demanded units. In the hard-limit replay an order is truncated to $\text{max_storage_units}_i$ minus the inventory position, and no order is placed if the truncated quantity is zero.

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The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this

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795 **Data and Code Availability**

796 The raw operational data and material-level artifacts cannot be publicly
797 released because they contain commercially sensitive material, supplier
798 and planning information. A confidentiality-screened release package
799 containing analysis code, the anonymized variable schema, aggregate
800 evaluation results, manuscript sources and prompt files is available at
801 github.com/DressPD/inventor_public. A versioned Zenodo archive of
802 the release is available at <https://doi.org/10.5281/zenodo.22262052>.
803 Supplementary File S1 contains the system prompt used for all runs; the
804 prompt file, the runner code and, for each call, a hash of the input, the
805 UTC time, the session identifier and the raw response are in the release.
806 The model identifier and inference settings are not exposed by the platform
807 and are operator-declared (Anthropic Claude Sonnet 4.6).

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814 Declaration of Generative AI and AI-Assisted Technologies in the 815 Writing Process

816 During manuscript preparation and revision, the author used OpenAI
817 GPT-5.6 and Anthropic Claude models through the OpenCode coding assis-
818 tant for language editing, restructuring, consistency checks, citation verifica-
819 tion against Crossref metadata, LaTeX proofing, and writing and executing
820 the analysis scripts that produce the evidence tables. These assistants are
821 distinct from the Claude Sonnet 4.6 deployment evaluated in the study. The
822 author specified the analyses, reviewed and edited all content and results,
823 and takes full responsibility for the content. No generative AI created or
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